

Grade 3-4 Severe symptoms limiting self care ADL	Permanently discontinue nivolumab treatment.	Obtain neurology opinion. Treat symptoms as per local guidelines. Steroid treatment as per local policy (recommended dose 1-2mg/kg/day IV methylprednisolone or equivalent dose of oral steroids). Prophylactic antibodies for opportunistic infections. If improves to G2: Taper steroids over at least 1 month. If worsens or atypical presentation: Consider IVIg or other immunosuppressive therapies as per local guidelines.
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8.5.1.8 Management of other immune-related adverse reactions

Adequate evaluation should be performed to confirm aetiology and exclude other causes.

See summary of product characteristics for detailed guidance on management of individual toxicities.

CTCAE v5 grade	Action – trial treatment	Recommended management of toxicity
Grade 1	Continue nivolumab treatment at 240mg.	Treatment for symptoms as per local policy.
Grade 2 (first occurrence)	Delay nivolumab treatment Restart nivolumab at 240mg if/when returns to $\leq$ G1. If steroids given, patient must have been weaned to $\leq$ 10mg per day (or equivalent) before restarting treatment. Maximum permitted delay 4 weeks.	Treat symptoms as per local guidelines. Steroid treatment as per local policy; taper prior to restarting nivolumab.
Grade 3 (first occurrence – except myocarditis – see below)	Delay nivolumab treatment. Restart nivolumab at 240mg if/when returns to $\leq$ G1. If steroids given, patient must have been weaned to $\leq$ 10mg per	Treat symptoms as per local guidelines. Steroid treatment as per local policy; taper prior to restarting nivolumab.

	day (or equivalent) before restarting treatment. Maximum permitted delay 4 weeks.	
Grade 3 myocarditis	Permanently discontinue nivolumab treatment.	Refer for specialist opinion. Manage as per local policy.
Grade 4 or recurrent grade 3; Persistent grade 2 or 3 despite intervention; Inability to reduce corticosteroid dose to 10mg prednisolone per day or equivalent	Permanently discontinue nivolumab treatment.	Treat symptoms as per local guidelines. Steroid treatment as per local policy. Seek specialist opinion.

8.5.2. Recommendations for Treatment of Nivolumab Related Infusion Reaction

As nivolumab is a fully humanised immunoglobulin, it is unlikely to be immunogenic and induce an infusion or hypersensitivity reaction. However, if such a reaction were to occur, it might manifest with fevers, chills, rigors, rash, headache, arthralgia, hypo- or hypertension, bronchospasm. All grade 3 or grade 4 infusion reactions should be reported as an SAE within 24 hours (see section 12.4.4 for details of reporting).

For grade 1 symptoms (Mild reaction, infusion interruption not indicated)

- Remain at bedside and monitor subject until recovery from symptoms.
- Premedication is recommended prior to subsequent infusions. Administration of 10mg IV chlorphenamine and paracetamol 1g PO at least 30 minutes prior to infusion is recommended, but local policies may be followed.

For grade 2 symptoms (Moderate reaction requires therapy or infusion interruption but responds promptly to symptomatic management)

- Stop the nivolumab infusion
- Give supportive care as per local policies. IV infusion of 0.9% sodium chloride solution for injection, and treatment with 1g paracetamol PO, 10mg IV chlorphenamine is recommended. Consider bronchodilator therapy and corticosteroids, up to 25mg IV hydrocortisone is recommended.
- Restart the infusion at 50% of the original infusion rate when symptoms resolve. If no further complications occur after 30 minutes, the rate may be increased to 100% of the original infusion rate. Monitor subjects closely.
- Premedication is recommended prior to subsequent infusions. Use of paracetamol, chlorphenamine and hydrocortisone prior to infusion is recommended, but local policies may be followed.

For grade 3 or 4 symptoms (Grade 3: not rapidly responsive to symptomatic management; recurrence of symptoms following initial improvement) (Grade 4 life threatening; pressor or ventilator support indicated)

- Stop the nivolumab infusion
- Give supportive care as per local policies. An IV infusion of 0.9% sodium chloride solution for injection and treatment with paracetamol, chlorphenamine, bronchodilators (as described for grade 2 reactions above), and 100mg IV methyl-prednisolone or equivalent is recommended.
- Follow institutional guidelines for anaphylaxis if indicated and involve intensive care/critical care outreach teams at an early stage.

Tumour lysis risk

Nivolumab can be associated with tumour lysis syndrome, and patients should be risk-assessed and managed as per local protocols for patients undergoing chemotherapy.

8.5.3. Recommendations for management of pDILI

The diagnostic criteria for pDILI are:

- AST or ALT >3 x ULN
- and**
- Total bilirubin >2 x ULN, without initial findings of cholestasis (elevated serum alkaline phosphatase)
- and**
- No other immediately apparent causes of transaminase elevation and hyperbilirubinemia, including but not limited to, viral hepatitis, pre-existing chronic or acute liver disease, or the administration of other drug(s) known to be hepatotoxic

There is no specific management for pDILI. If a patient presents with abnormal liver function tests as outlined above, further investigation should be carried out to establish whether the patient has pDILI or autoimmune liver disease.

- Nivolumab should be permanently discontinued.
- Treatment should be given as clinically indicated.
- If autoimmune liver disease is suspected, this should be managed as per section 8.5.1.3 above

8.6. Management of Overdoses, Trial treatment error or Occupational Exposure

Overdose

Administration of a quantity of a trial treatment, either per administration or cumulatively, which is in excess of the protocol specified dose. The dose can either be evaluated as overdose by the trial team at site or by the Sponsor upon review.

No specific information is available on the management of nivolumab overdose. The patient should be carefully monitored and supportive care instituted as appropriate.

Any overdoses require expedited reporting to UCL CTC as an SAE (see section [12.2.2](#) (Reporting of serious adverse events) for reporting procedures).

Any overdoses must also be reported as an incident according to the instructions in section 13.1 (Incident reporting).

See also section 12.4.3 (Overdoses).

Trial treatment error

Any unintentional error in prescribing, dispensing, or administration of a trial treatment while in the control of a healthcare professional or consumer. The error can be identified either by the trial team at site or by the Sponsor upon review.

Include details from IB/SPC on the clinical management of trial treatment errors, where available. Trial Treatment errors should be reported on in incident report (see section [13.1](#)). Any adverse events resulting from a medication error should be reported as an SAE (see section [12.2.2](#) for reporting procedures).

Occupational exposure

Exposure to a trial treatment as a result of one's professional or non-professional occupation. Occupational exposure should be reported on an incident report form (see section [13.1](#)).

8.7. Supportive Care

No specific supportive medication is advised with nivolumab. Recommended supportive care for patients with autoimmune complications or infusion reactions are set out in section 8.4.

Anti-emetics, anaglesics, antibiotics, electrolyte and fluid replacement, anti-coagulants and anti-platelet agents may be given in accordance with local policies and as clinically indicated.

Medications used to treat pre-existing co-morbidities can be continued, with the exception of immune-suppressants (as set out in section 8.8). Supportive/prophylactic medications recommended to continue post previous chemotherapy regimes, e.g. acyclovir, may also be continued.

GCSF may be given as per local policy for neutropenia persisting for greater than 7 days.

Any supportive medications given should be clearly documented in the patient notes.

8.8. Contraindications

According to the IB for nivolumab, no drugs are contraindicated during nivolumab treatment.

Any other systemic anti-cancer therapy or radiotherapy is prohibited while the patient is receiving trial treatment. Steroids should be weaned down to ≤ 10 mg/day (prednisolone; or the equivalent dose of other steroids) within 7 days prior to starting trial treatment.

8.9. Pharmacy Responsibilities

All pharmacy aspects of the trial at participating sites are the responsibility of the PI, who may delegate this responsibility to the local pharmacist or other appropriately qualified personnel, who will be the Pharmacy Lead. The delegation of duties must be recorded on the site staff delegation log.

Nivolumab supplied for the ANIMATE trial are for eligible ANIMATE trial patients only and must not be used outside the context of this protocol.

8.9.1. Temperature Excursions

All temperature excursions outside the storage conditions specified in the IB/Summary of Drug Arrangements/labels must be reported to UCL CTC as per the 'Pharmacy Procedure for Reporting Temperature Excursions' (see Pharmacy Site File)

Upon identifying an excursion:

- all affected trial stock must be quarantined IMMEDIATELY
- a 'Notification of Temperature Excursion' form must be completed and e-mailed to ctc.excursions@ucl.ac.uk or faxed to +44 (0)20 7679 9861.

Please note that UCL CTC must be informed immediately if a patient has been administered drug affected by a temperature excursion.

8.9.2. IMP accountability

The Pharmacy Lead must ensure that appropriate records are maintained.

These records must include accountability for each drug including: receipt, dispensing, reconciliation and destruction of unused medication (on sponsor authorisation). Accountability forms will be supplied, and must be used, unless there is prior agreement from UCL CTC to use alternative in-house records.

Copies of completed drug accountability logs must be submitted to UCL CTC for all trial patients upon request. Also refer to section 14.1 (Central Monitoring).

8.10. Clinical Management after Treatment Discontinuation

Clinical management following treatment discontinuation will be at the treating clinician's discretion.

If a patient discontinues trial treatment early, they will remain on trial for follow up purposes unless they explicitly withdraw consent. Also refer to sections 9 (Assessments) and 15 (Withdrawal of Patients) for further details regarding treatment discontinuation, patient withdrawal from trial treatment and withdrawal of consent to data collection.

8.11. Drug Provision after the End of the Trial

BMS will supply sufficient nivolumab for all eligible patients to receive up to 8 cycles of nivolumab. Patients will not continue treatment beyond 8 cycles, and there are no arrangements in place for supply following the end of the trial.

9. ASSESSMENTS

Please also see Schedule of Events table in Appendix 2

9.1. Pre-registration Assessments

Registration will take place before, during or within 14 days of completing the first 2 cycles of first or second line salvage treatment (4 cycles if treated with brentuximab vedotin). Patients must give written informed consent **before** any trial specific screening investigations may be carried out. The following assessments or procedures are required to evaluate the suitability of patients for the trial and must be carried out within 28 days prior to registration unless otherwise stated:

- Medical history
- Pregnancy test (serum or urine; women of childbearing potential; within 24 hours prior to registration)

Following study registration, the tumour block from the patient's lymphoma biopsy (preferably biopsy performed at time of relapse, or if unavailable the biopsy from their initial diagnosis) must be retrieved from the hospital's pathology archive and sent to the central laboratory (HMDS, Leeds). See section 10.1 and the Laboratory Manual for details.

9.2. Assessment of eligibility for nivolumab treatment

Following first or second line salvage, the patient will have assessments to verify they are eligible for treatment with Nivolumab.

The following investigations will be carried out within 18 days of cycle 2 first or second line salvage chemotherapy (cycle 4 if being treated with brentuximab vedotin) ± 3 days:

- PET-CT scan (PET0; to be carried out as per trial requirements on an approved scanner; images to be sent to the PET core lab for central review – see the trial Imaging Manual for details).
- Contrast enhanced CT scan (ceCT). If feasible, this should be performed at the same imaging session as the PET-CT. If performed at the same session, the ceCT scan should be performed after the low dose PET-CT (see trial Imaging Manual for details).

The following investigations will be carried out up to 3-6 weeks after last administration of cycle 2 first or second line salvage chemotherapy (cycle 4 if being treated with brentuximab vedotin) if PET-CT was positive (Deauville 4-5):

- Medical history
- Clinical examination
- Weight
- Full blood count with differential (must be performed at least 3 weeks post completion of salvage)
- Biochemistry (ALT/AST, total bilirubin, alkaline phosphatase, urea, creatinine, calcium, magnesium, sodium, potassium, LDH, albumin, urate, glucose) (must be performed at least 3 weeks post completion of salvage)
- Creatinine clearance (calculated by Cockcroft-Gault method)
- Tests for autoimmune conditions (amylase, lipase, thyroid function tests, adrenocorticotrophic hormone (ACTH))
- Electrocardiogram (ECG)
- Echocardiogram (if clinically indicated e.g. abnormalities on ECG)

- Lung function test
- ECOG performance status
- Hepatitis B & C serology:
 - hepatitis B surface antigen (HBsAg)
 - hepatitis B surface antibody (HBsAb) – [if standard locally](#)
 - hepatitis B core antibody (HBcAb)
 - ~~hepatitis B antibodies or~~ hepatitis B DNA (HBV DNA) [IF hepatitis B core antibody positive](#)
 - hepatitis C antibody [yes](#) or hepatitis C DNA (HCV DNA) [\(local standard test to be used\)](#)

The following investigation is to be carried out within 24 hours prior to confirming eligibility for treatment:

- Pregnancy test (serum or urine; women of childbearing potential)

The table below summarises the permitted timeframes for performing eligibility assessments. NB treatment with nivolumab must not be started until eligibility for trial treatment has been confirmed.

Week post first or second line salvage	1	2	3	4	5	6
PET CT ¹						
Other treatment eligibility tests ²						
Notes 1 = PET CT to be performed 18 days after last administration of salvage treatment ± 3 days 2 = FBC and biochemistry must be performed at least 3 weeks after after last administration of cycle 2 salvage treatment (or cycle 4 of brentuximab vedotin) to allow time for count recovery. For WOCBP, a pregnancy test is required within 24 hours prior to starting nivolumab.						

9.3. Assessments prior to cycle 1 of Nivolumab

The following investigations are to be performed within 24 hours prior to starting Nivolumab:

- Pregnancy test (urine or serum; WOCBP only)

The following investigations are to be performed within 3 days prior to starting Nivolumab:

- Clinical examination, including documentation of baseline adverse events
- Full blood count with differential*
- Biochemistry (ALT/AST, total bilirubin, alkaline phosphatase, urea, creatinine, calcium, magnesium, sodium, potassium, LDH, albumin, urate, glucose) *
- Thyroid function tests (TSH & Free T4; Free T3 to be performed if TSH/Free T4 abnormal)
- [Test for autoimmune conditions \(Amylase, lipase, ACTH\)](#)
- Blood samples [s](#) to be taken and sent to the central laboratory at the [Weatherall Institute of Molecular Medicine, Oxford](#) (see section 10.2 and Laboratory Manual for details)

* Do not need to be repeated if investigations for confirmation of eligibility (see section 9.2) were performed within 3 days prior to starting Nivolumab

The following investigations must be carried out on the day of nivolumab administration:

- Oxygen saturation measured by pulse oximetry at rest

9.4. Assessments during Treatment

During treatment the patient should be seen every 2 weeks and the following assessments performed within 3 days prior to the start of each cycle:

- Clinical examination
- Full blood count with differential
- Biochemistry (ALT/AST, total bilirubin, alkaline phosphatase, urea, creatinine, calcium, magnesium, sodium, potassium, LDH, albumin, urate, glucose)
- Tests for autoimmune conditions (amylase, lipase, ACTH)
- Cycles 3, 5 & 7: pregnancy test (urine or serum; WOCBP only)
- Cycle 4 and cycle 7: Thyroid function tests (TSH & Free T4; Free T3 to be performed if TSH/Free T4 abnormal)
- Adverse event assessment

The following investigations must be carried out on the day of nivolumab administration:

- Oxygen saturation measured by pulse oximetry at rest

Blood samples for translational research

- Blood samples are to be taken and sent to the central laboratory at the Weatherall Institute of Molecular Medicine, Oxford within 3 days prior to the start of cycles 2, 4, 6 and 8, and at the 1 month post-treatment visit (see section 10.2 and Laboratory Manual for more details)

Scanning required after cycle 4

- PET-CT scan to be carried out on cycle 4, day 11-13 as per trial requirements on an approved scanner, and images sent to the PET core lab for central review (PET4; see trial Imaging Manual for details).
- Contrast enhanced CT scan (ceCT). If feasible, this should be performed at the same imaging session as the PET-CT. If performed at the same session, the ceCT scan should be performed after the low dose PET-CT (see trial Imaging Manual for details).

9.5. Assessments on Completion of Trial Treatment

For patients who withdraw from trial treatment for any reason during cycles 1-3 of nivolumab:

- Lung function test (to be performed as soon as possible after decision to stop trial treatment, and before the next line of treatment begins)

For patients in CMR (Deauville 1-3) or with progressive disease after 4 cycles of nivolumab

- Lung function test (to be performed as soon as possible after receiving the PET-CT central review result)

If a biopsy is performed to confirm progressive disease, please send the block to HMDS, Leeds (see section 10).

For patients who withdraw from trial treatment for any reason during cycles 5-7:

- PET-CT scan to be carried out at least 11-13 days after last trial treatment administration, as per trial requirements on an approved scanner, and images sent to the PET core lab for central review (see trial Imaging Manual for details).
- Lung function test (to be performed as soon as possible after decision to stop trial treatment, and before the next line of treatment begins)

For patients who complete all 8 cycles of trial treatment:

- PET-CT scan to be carried out on cycle 8, day 11-13 as per trial requirements on an approved scanner, and images sent to the PET core lab for central review (PET8; see trial Imaging Manual for details).
- Contrast enhanced CT scan (ceCT). If feasible, this should be performed at the same imaging session as the PET-CT. If performed at the same session, the ceCT scan should be performed after the low dose PET-CT (see trial Imaging Manual for details).
- Repeat lymph node biopsy (if patient is PET positive and has consented for a repeat biopsy). Block to be sent to central laboratory (HMDS) (see section 10.1 and Laboratory Manual for details)
- Lung function test (to be completed 10 days after last nivolumab administration ± 2 days)

9.6. Assessments During Follow Up

Patients will be followed up for a minimum of 3 years. All efforts should be made by the Site to contact the patient's GP to assess their condition, if a patient fails to attend a clinic or cannot be followed up at site.

9.6.1. Patients who are ineligible for Nivolumab

Patients will be seen as follows:

- 3 months after PET0 scan (i.e. scan performed at end of first or second line salvage)
- 6 months after PET0 scan
- 9 months after PET0 scan
- 12 months after PET0 scan
- Annually thereafter

The following investigations will be performed:

- Remission status assessment (investigations as per local protocols).

9.6.2. Patients who receive nivolumab**Follow up in year 1 post treatment**

Patients will be seen approximately once a month for the first 3 months, and every 3 months thereafter.

The following tests will be performed **1 month, 2 months and 3 months** post treatment (30, 60 & 90 days after the last dose of Nivolumab respectively (± 3 days)):

- Clinical examination
- Full blood count with differential
- Biochemistry (ALT/AST, total bilirubin, alkaline phosphatase, urea, creatinine, calcium, magnesium, sodium, potassium, LDH, albumin, urate, glucose)

- Tests for autoimmune conditions (amylase, lipase, ACTH)
- Thyroid function tests (TSH & Free T4; Free T3 to be performed if TSH/Free T4 abnormal)
- Adverse event assessment
- Pregnancy test (urine or serum; WOCBP only)
- Remission status assessment (investigations as per local policy) – 3 month timepoint only

Blood samples for translational research

- Blood samples are to be taken and sent to the central laboratory at the Weatherall Institute of Molecular Medicine, Oxford at the 1 month post-treatment visit (30 days post last dose of nivolumab (± 3 days); see section 10.2 and Laboratory Manual for more details)

The following tests will be performed at **months 6, 9 and 12** post treatment:

- Clinical examination
- Full blood count with differential
- Biochemistry (U&E, LFTs)
- Remission status assessment (investigations as per local policy)
- Assessment for late toxicity of nivolumab

In addition, at **month 12** post treatment, the following test will be carried out:

- Lung function test

Follow up from year 2 post treatment onwards

Patients will be seen annually. The following investigations will be performed:

- Remission status assessment (investigations as per local protocols)
- Assessment for late toxicity of nivolumab

9.7. Assessments at time of Disease Progression

If disease progression is suspected, investigations should be carried out as per local protocols. If progression is confirmed, a disease progression urgent event form must be submitted within 72 hours of becoming aware (see section 12.5.2 for more information).

Samples for translational research

- If a repeat biopsy is performed at the time of relapse, the block should be sent to the central laboratory (HMDS) (see section 10.1 and Laboratory Manual for details)

9.8. Follow up Assessments after Disease Progression

Following disease progression, patients will be seen annually. The following investigations will be performed:

- Remission status assessment (investigations as per local protocols)
- Assessment for late toxicity of nivolumab

N.B. confirmed disease progressions must be reported as urgent events within 72 hours of becoming aware (see section 12.5 (Urgent Events))

10. EXPLORATORY BIOLOGICAL STUDIES

Biological studies are an integral part of the trial and as such are not optional but will be built into the main study consent (apart from an optional biopsy of residual PET avid lesions). We will be collecting two types of sample: formalin fixed paraffin embedded tissue and blood.

There are ~~two~~three central laboratories for the trial. The Haematological Malignancy Diagnostic Service (HMDS) in Leeds will be responsible for central pathology review using immunohistochemistry (IHC), gene expression profiling (GEP) and fluorescent in situ hybridisation (FISH) studies. The Weatherall Institute of Molecular Medicine, Oxford will be carrying out biomarker studies on blood samples using flow cytometry. The MRC-University of Glasgow Centre for Virus Research will perform TARC analysis on blood samples.

Sites must keep a record of all samples sent to central laboratories for the trial, and ensure they update the sample tracking website when samples are sent. –See the Laboratory Manual for more details of how to track samples.

10.1. Formalin fixed paraffin embedded blocks

Formalin fixed paraffin embedded (FFPE) tissue will be sent to HMDS, Leeds from the referring hospitals. This will preferably be from a lymph node biopsy taken at first relapse. In the absence of this material, the initial diagnostic sample will be acceptable. It should preferably be a biopsy from an involved lymph node, or solid tumour mass. In the absence of this, any diagnostic material is acceptable (e.g. bone marrow trephine biopsy with involvement by Hodgkin). The block should be sent to HMDS, Leeds. See the Laboratory Manual for details of how to send and log FFPE samples. The consent for laboratory studies will either accompany the tissue sample or a copy of the consent form will be faxed to HMDS. Laboratory processing of any samples will not occur until patient consent has been confirmed.

In addition, an optional biopsy will be performed of any PET avid residuum after nivolumab treatment. This will also be formalin fixed and paraffin embedded and sent to HMDS, Leeds.

If, following trial treatment, a biopsy is taken to confirm disease progression, the block should be sent to HMDS, Leeds.

3µm sections of the tissue block will be taken and immunohistochemistry performed. A panel of markers will be assessed for each case including: CD30, IRF-4, PAX-5, CD20, CD3, OCT-2, BCL-6, CD79, CD15 and LMP-1. Immunostaining for PD1, PD-L1 and PD-L2 will also be performed in most cases.

Sections of the block will also be prepared for FISH studies. The FISH studies conducted will include those for 9p24 abnormalities as these have previously been reported to be highly predictive of outcome (Roemer et al 2016). Probes will include CD274/PD-L1, PDCD1LG2/PD-L2, CEP9 and REL. Fifty Reed-Sternberg cells per slide will be reviewed and the FISH signals defined as disomic, polysomic, relative copy gain or amplified.

GEP will be performed using either or both of the HTG Immunopanel and the Affymetrix Human Transcriptome Array, which is a whole transcriptome array. These are two gene expression platforms currently being used in HMDS with bioinformatic support from within HMDS and in collaboration with the University of Leeds.

Once analysis is complete, blocks can be returned to sites at their request.

10.2. Peripheral blood samples

Research peripheral blood samples will also be collected as follows:

Timepoint	Sample required
Within 3 days prior to start of cycle 1	50ml peripheral blood in EDTA <u>5ml peripheral blood in serum gel tube</u>
Within 3 days prior to start of cycle 2	20ml peripheral blood in EDTA <u>5ml peripheral blood in serum gel tube</u>
Within 3 days prior to start of cycle 4	20ml peripheral blood in EDTA <u>5ml peripheral blood in serum gel tube</u>
Within 3 days prior to start of cycle 6	20ml peripheral blood in EDTA <u>5ml peripheral blood in serum gel tube</u>
Within 3 days prior to start of cycle 8	20ml peripheral blood in EDTA <u>5ml peripheral blood in serum gel tube</u>
1 month post treatment follow up visit	20ml peripheral blood in EDTA <u>5ml peripheral blood in serum gel tube</u>

Samples should be sent by first class mail to the central laboratory at the Weatherall Institute of Molecular Medicine, Oxford. No processing of the sample is required at site prior to sending. Samples taken on a Friday should be refrigerated over the weekend and sent on the next working day. Please refer to the Laboratory Manual for more details of how to send and log blood samples.

EDTA samples will be analysed at the Weatherall Institute of Molecular Medicine, Oxford. Immunophenotyping by multicolour flow panels and/or CyTOF (depending on cell recovery) will be performed on the blood to identify immune signatures prior to, during, and after nivolumab treatment. This may give important predictive and pharmacodynamic information. Panels for myeloid and T cell markers will be used. For patients with circulating activated T cells, these will be sorted and analysed functionally. Sequencing of TCR will also be performed. Any leftover material will be stored in a biobank for future use in ethically approved research, subject to patient consent.

Serum samples will be processed by the Weatherall Institute and frozen in the first instance. They will then be transferred to the MRC-University of Glasgow Centre for Virus Research where measurement of TARC (CCL17) levels will be performed. It is anticipated that samples will be transferred in one batch once all patients have completed trial treatment, and all TARC analysis will be performed retrospectively. Any leftover material following this analysis will be stored in a biobank for future use in ethically approved research, subject to patient consent.

Peripheral blood samples, or any products produced from these, will not be returned to sites.

10.3. Agreements and accreditations

The trial consent form will include provision for samples to be sent to, and stored at, the HMDS, and the Weatherall Institute of Molecular Medicine, Oxford and the MRC-University of Glasgow Centre for Virus Research. The ethical approval of the trial will cover storage and processing of

samples at the HMDS, ~~and~~ the Weatherall Institute of Molecular Medicine, Oxford and the MRC-University of Glasgow Centre for Virus Research in accordance with the protocol.

HMDS is a CPA accredited laboratory with UKAS accreditation to ISO15189 standards.

UCL CTC will hold laboratory agreements with the HMDS, ~~and~~ the Weatherall Institute of Molecular Medicine, Oxford and the MRC-University of Glasgow Centre for Virus Research.

11. DATA MANAGEMENT AND DATA HANDLING GUIDELINES

Data will be collected from sites on version controlled case report forms (CRFs) designed for the trial and supplied by UCL CTC. Data must be accurately transcribed onto trial CRFs and must be verifiable from source data at site. Examples of source documents are hospital records which include patient's notes, laboratory and other clinical reports etc.

Where copies of supporting source documentation (e.g. autopsy reports, pathology reports, CT scan images etc.) are being submitted to UCL CTC, the patient's trial number must be clearly indicated on all material and any patient identifiers removed/blacked out prior to sending to maintain confidentiality.

Please note that, for this trial, patients must consent to their NHS/CHI being supplied to UCL CTC. This is required in order to allow for follow up via national registries where needed.

11.1. Completing Case Report Forms

All CRFs must be completed and signed by staff who are listed on the site staff delegation log and authorised by the PI to perform this duty. The PI is responsible for the accuracy of all data reported in the CRF.

All entries must be clear, legible and written in ball point pen. Any corrections made to a CRF at site must be made by drawing a single line through the incorrect item ensuring that the previous entry is not obscured. Each correction must be dated and initialed. Correction fluid must not be used.

The use of abbreviations and acronyms should be avoided.

11.2. Missing Data

To avoid the need for unnecessary data queries CRFs must be checked at site to ensure there are no blank fields before sending to UCL CTC (unless it is specifically stated that a field may be left blank). When data are unavailable because a measure has not been taken or test not performed, enter "ND" for not done. If an item was not required at the particular time the form relates to, enter "NA" for not applicable. When data are unknown enter the value "NK" (only use if every effort has been made to obtain the data).

11.3. Timelines for Data Return

CRFs must be completed at site and returned to UCL CTC as soon as possible after the relevant visit and within the following timeframes:

- Registration CRF – to be faxed/emailed to UCL CTC in real time
- Post salvage CRF – to be faxed/emailed to UCL CTC in real time
- Treatment CRFs – to be sent to UCL CTC within 2 weeks of the end of each cycle
- Treatment Summary CRF – to be sent to UCL CTC within 2 weeks of the decision to stop treatment (with the Treatment CRF for the last cycle given)
- Follow up CRFs – to be sent to UCL CTC within 4 weeks of the patient being seen
- SAE forms – to be faxed or emailed to UCL CTC within 24 hours of becoming aware of the event

- Pregnancy report forms – to be faxed or emailed to UCL CTC within 24 hours of becoming aware of the pregnancy
- Post-allogeneic transplant urgent event form – to be faxed or emailed to UCL CTC within 72 hours of becoming aware of the event
- Disease progression form – to be faxed or emailed to UCL CTC within 72 hours of becoming aware of confirmed disease progression
- Death form – to be faxed or emailed to UCL CTC within 24 hours of becoming aware of the death.
- New treatment form - to be faxed or emailed to UCL CTC with the next due Follow up CRF.
- Transfer of care form – to be faxed or emailed to UCL CTC within 2 weeks of the transfer of care to another site.
- Lost to follow up form - to be faxed or emailed to UCL CTC within 1 month of becoming aware that the patient is lost to follow up.

Sites that persistently do not return data within the required timelines may be suspended from recruiting further patients into the trial by UCL CTC and subjected to a 'for cause' monitoring visit. See section 14.2 ('For Cause' On-Site Monitoring) for details.

11.4. Data Queries

Data arriving at UCL CTC will be checked for legibility, completeness, accuracy and consistency, including checks for missing or unusual values. Data Clarification Requests will be sent to the data contact at site. Further guidance on how data contacts should respond to data queries can be found on the Data Clarification Request forms.

12. PHARMACOVIGILANCE

12.1. Definitions

The following definitions have been adapted from Directive 2001/20/EC, ICH E2A “Clinical Safety Data Management: Definitions and Standards for Expedited Reporting” and ICH GCP E6.

Adverse Event (AE)

Any untoward medical occurrence in a patient treated on a trial protocol, which does not necessarily have a causal relationship with a trial treatment. An AE can therefore be any unfavourable and unintended sign (including an abnormal laboratory finding), symptom or disease temporally associated with the use of a trial treatment, whether or not related to that trial treatment. See section 12.2.1 for AE reporting procedures.

Adverse Reaction (AR)

All untoward and unintended responses to a trial treatment related to any dose administered. A causal relationship between a trial treatment and an adverse event is at least a reasonable possibility, i.e. the relationship cannot be ruled out.

Serious Adverse Event (SAE) or Serious Adverse Reaction (SAR)

An adverse event or adverse reaction that at any dose:

- Results in death
- Is life threatening (the term “life-threatening” refers to an event in which the patient was at risk of death at the time of the event. It does not refer to an event that hypothetically might have caused death if it were more severe)
- Requires in-patient hospitalisation or prolongs existing hospitalisation
- Results in persistent or significant disability/incapacity
- Is a congenital anomaly or birth defect
- Is otherwise medically significant (e.g. important medical events that may not be immediately life-threatening or result in death or hospitalisation but may jeopardise the patient or may require intervention to prevent one of the other outcomes listed above)

See section 12.2.2 for SAE reporting procedures.

Suspected Unexpected Serious Adverse Reaction (SUSAR)

An adverse event meeting the following criteria:

- Serious – meets one or more of the serious criteria above
- Related – assessed by the local investigator or sponsor as causally related to one or more elements of the trial treatment
- Unexpected – the event is not consistent with the applicable reference safety information (RSI)

See section 12.3 for reporting procedures for these events.

Adverse event of special interest (AESI)

An AE that is of particular interest to the Trial Management Group, even if it occurs outside the standard AE reporting timeframes for the trial. They are to be reported on an SAE form, regardless of their seriousness. The AEs of special interest for this trial are:

AESIs to be reported as SAEs

- Autoimmune diarrhoea / colitis
- Autoimmune pneumonitis
- Autoimmune hepatitis
- Autoimmune dermatitis
- Immune related endocrinopathies
- Autoimmune renal toxicity
- Autoimmune neurological toxicity
- Autoimmune uveitis
- Potential drug induced liver injury (pDILI)
- Overdoses (regardless of whether patient experiences any adverse events as a result)
- Grade 3 or 4 infusion reactions to nivolumab

See section 12.4 for reporting procedures for these events.

Urgent event

Urgent events for this trial are:

- Grade 3/4 graft versus host disease (patients who have allogeneic transplant only)
- Hyperacute graft versus host disease (patients who have allogeneic transplant only)
- Steroid-responsive febrile syndrome (patients who have allogeneic transplant only)
- Disease progression
- Death

See section 12.5 for reporting procedures for these events.

Overdose, trial treatment error, or occupational exposure

Refer to section ☐ for details on reporting of these events. Additionally refer to section 12.4.3 for details of reporting overdoses.

12.2. Reporting Procedures

Adverse Event Term

An adverse event term must be provided for each adverse event. Wherever possible a valid term listed in the Common Terminology Criteria for Adverse Events (CTCAE) v5.0 should be used. This is available online at:

<http://evs.nci.nih.gov/ftp1/CTCAE/>

Severity grade

Severity grade of each adverse event must be determined by using CTCAE v5.0. This is available online at:

<http://evs.nci.nih.gov/ftp1/CTCAE/>

Causality

The relationship between the treatment and an adverse event will be assessed.

For AEs, the local PI or designee will assess whether the event is causally related to trial treatment.

For SAEs, a review will also be carried out by the Sponsor's delegate.

Causal relationship to each trial treatment must be determined as follows:

- Related (reasonable possibility) to a trial treatment
- Not related (no reasonable possibility) to a trial treatment

UCL CTC will consider events evaluated as related to be adverse reactions.

Responsibility for reporting safety events following transfer of care

In the event of transfer between trial sites

UCL CTC must be informed if a patient's care is transferred to another trial site (see section 15.4 – Transfer of Care). The trial site caring for the patient at the time of awareness of AE, SAE, AESI, Urgent Event onset or pregnancy is responsible for completing and sending relevant reports to UCL CTC. When the patient's care is transferred, the referring site must provide sufficient information about the patient's trial participation (i.e. a copy of the patient's completed CRFs up to the date of transfer) to the new site to allow them to complete SAE reports.

In the event of transfer between a trial site and a non-trial site

The trial site will remain responsible for completing and sending AE, SAE, AESI Urgent Event and pregnancy reports to UCL CTC. If a patient is discharged to the care of a non-trial site, the PI/study team at the trial site should ensure that they are sent regular updates on the patient's progress and notified of any admissions or significant medical developments (including relapses and deaths), in order that the site's reporting responsibilities can be fulfilled.

12.2.1. Reporting of Adverse Events (AEs)

All adverse events that occur between the start of nivolumab treatment and 3 months post last nivolumab administration or commencement of next treatment for lymphoma (whichever is earlier) must be recorded in the trial CRFs.

The reporting period has been selected as the patients are followed up at 1, 2, 3 months, then moving to 3 monthly follow up schedule. It is anticipated that most patients will have proceeded to further lymphoma treatment by the 3 month timepoint.

Adverse events meeting the definition of a Serious Adverse Event (SAE) or Adverse Event of Special Interest (AESI) must also be reported to UCL CTC using the trial specific SAE Report. Also refer to section 12.2.2 (Reporting of Serious Adverse Events (SAEs)) and section 12.4 (Adverse Events of Special Interest). Adverse events meeting the definition of an Urgent Event must also be reported on an Urgent Event form (also refer to section 12.5 Urgent Events).

Pre-existing conditions do not qualify as adverse events unless they worsen.

12.2.2. Reporting of Serious Adverse Events (SAEs)

All SAEs that occur between the start of nivolumab treatment and 5 months post last nivolumab administration or commencement of next treatment for lymphoma (whichever is earlier), including

stem cell transplant (**or after this date if the site investigator feels the event is related to nivolumab**) must be submitted to UCL CTC.

The reporting timeframe takes into account the potential risk of delayed reactions from nivolumab (5 times half life, approximately 5 months), but is balanced against the fact that patients are expected to proceed to further lymphoma treatment after nivolumab, which will have a similar (or worse) toxicity profile compared to nivolumab. The different reporting requirements for AESIs (see section 12.4 below) allows for continued reporting of important nivolumab-specific reactions (such as immune-related events), even after the start of a new lymphoma treatment.

SAEs must be reported by fax or email within **24 hours** of observing or learning of the event, using the trial specific SAE Report. All sections on the SAE Report must be completed. If the event is **not being reported within 24 hours** to UCL CTC, the circumstances that led to this must be detailed in the SAE Report to avoid unnecessary queries.

The following AESIs must be reported on the SAE report form:

- Autoimmune diarrhoea / colitis
- Autoimmune pneumonitis
- Autoimmune hepatitis
- Autoimmune dermatitis
- Immune related endocrinopathies
- Autoimmune renal toxicity
- Autoimmune neurological toxicity
- Autoimmune uveitis
- Potential drug induced liver injury (pDILI)
- Overdoses (regardless of whether the patient experiences any adverse events as a result)
- Grade 3 or 4 infusion reactions to nivolumab

See section 12.4 for further information.

To summarise, the SAE reporting requirements for non-autoimmune events, based subsequent treatment, are as follows (see section 12.4 for details of reporting windows for autoimmune events):

Next Planned Treatment	Reporting window for non-autoimmune SAEs
No treatment	From first administration of nivolumab until 5 months post last dose of trial treatment. (N.B. if the patient relapses and starts new treatment within 5 months after completing nivolumab, SAEs will no longer be collected).
Further chemotherapy	From first administration of nivolumab until 5 months post last dose of trial treatment or first administration of next line of chemotherapy, whichever is sooner.
Autologous stem cell transplant	From first administration of nivolumab until 5 months post last dose of trial treatment or first administration of pre-transplant chemotherapy, whichever is sooner.
Allogeneic stem cell transplant	From first administration of nivolumab until 5 months post last dose of trial treatment or commencement of transplant conditioning, whichever is sooner.

Exemptions from SAE Report submission

For this trial, the following events are exempt from requiring submission on an SAE Report, but must be recorded in the relevant sections of the trial CRFs:

- Events that occur more than 5 months post last trial treatment administration unless:
 - considered to be a late effect of the trial treatment
 - it is a pregnancy related event (see section 12.8)
- Events that occur after the commencement of further treatment for lymphoma unless:
 - An autoimmune event classed as an AESI (see section 12.4)
- Disease progression (including disease related deaths, unless considered related to the trial treatment). N.B. disease progression and deaths from any cause are Urgent Events – see section 12.5)
- The following post-allogeneic transplant toxicities, which should be reported as urgent events (see section 12.5):
 - Grade 3/4 graft-vs-host disease (patients who have allogeneic transplant only)
 - Hyperacute graft-vs-host disease (patients who have allogeneic transplant only)
 - non-infectious febrile episodes requiring steroid therapy (including steroid-responsive febrile syndrome; patients who have allogeneic transplant only)
 - Sinusoidal obstruction syndrome/ veno-occlusive disease (patients who have allogeneic transplant only)

Please note that hospitalisation for elective treatment (including further treatment for lymphoma), palliative care, for logistical or socio-economic reasons does not qualify as an SAE.

Completed SAE Reports must be faxed or emailed to UCL CTC within 24 hours of becoming aware of the event

Fax: 020 7679 9861
Email: ctc.animate@ucl.ac.uk

SAE Follow-Up Reports

All SAEs will be followed up until resolution and until there are no further queries.

Sites must ensure any new and relevant information is provided promptly. If the event term changes or a new event is added, the causality must be re-assessed by an Investigator. If the event is not being reported within 24 hours to UCL CTC, the circumstances that led to this must be detailed in the SAE/SAR Report to avoid unnecessary queries.



SAE Processing at UCL CTC

On receipt of the SAE Report, UCL CTC will check for legibility, completeness, accuracy and consistency. Expectedness will be evaluated, to determine whether or not the case qualifies for expedited reporting, using the RSI (the approved Investigator Brochure (IB) for Nivolumab).

The CI, or their delegate (e.g. a clinical member of the TMG), may be contacted to review the SAE and to perform an evaluation of causality on behalf of UCL CTC. If UCL CTC has considered expectedness difficult to determine, the CI, or their delegate, will be consulted for their opinion at this time.

UCL CTC will submit all SAE Reports concerning patients who have received Nivolumab to BMS according to the timelines outlined in the agreement between UCL and BMS.

12.3. SUSARs

If the event is evaluated as a Suspected Unexpected Serious Adverse Reaction (SUSAR), UCL CTC will submit a report to the MHRA within the required timelines.

Wherever possible, evaluations of causal relationship by both the site and the Sponsor's clinical reviewer will be reported.

UCL CTC will submit all SUSAR reports relating to nivolumab to Bristol-Myers Squibb (BMS) according to the timelines outlined in the agreement between UCL and BMS.

Informing Sites of SUSARs

UCL CTC will inform all PIs of any SUSARs that occur on the trial. PIs will receive a quarterly line listing which must be processed according to local requirements.

12.4. Adverse events of special interest

All AESIs must be submitted to UCL CTC by fax or email within the timeframes outlined below, using the trial specific SAE Report. All sections on the SAE Report must be completed. If the event is **not being reported within the required timeframe** to UCL CTC, the circumstances that led to this must be detailed in the SAE Report to avoid unnecessary queries.

UCL CTC will submit all SAE Reports relating to AESIs in patients who have received Nivolumab to BMS according to the timelines outlined in the agreement between UCL and BMS.

12.4.1. Autoimmune events due to Nivolumab

The following adverse events of special interest for Nivolumab must be reported on the appropriate AE of Special Interest form within **24 hours of confirmed diagnosis**.

The following adverse events of special interest for Nivolumab must be reported on an SAE report form within **24 hours of confirmed diagnosis**. Please note that any signs and symptoms should be treated on their merits and reported as AEs/SAEs as applicable pending confirmation of diagnosis.

To summarise, the reporting requirements for **autoimmune events** based the treatment arm are as follows: (please note that the reporting window for autoimmune events differs to those for non-autoimmune SAEs; see section 12.2.2 for details).

Next Treatment	Reporting window for autoimmune toxicity
No treatment or further chemotherapy	From first administration of nivolumab until 5 months post last dose of trial treatment (or after this date if the investigator feels the event is related to nivolumab).
Autologous stem cell transplant	From first administration of nivolumab until 5 months post last dose of trial treatment (or after this date if the investigator feels the event is related to nivolumab).
Allogeneic stem cell transplant	From first administration of nivolumab until 5 months post last dose of trial treatment (or after this date if the investigator feels the event is related to nivolumab). Please note that colitis, dermatitis and hepatitis occurring after transplant day 0 should be reported as graft-versus-host disease in the first instance unless otherwise proven to be of autoimmune origin (see section 12.5 (Urgent Events) for details of reporting graft-versus-host disease).

Please note that confirmed autoimmune events must be reported to the CTC on an SAE report form even if the patient has received further treatment for their lymphoma within the 5 months after completing nivolumab.

Events will be collected as follows:

Event	Description See section 8.5 for detailed definitions	Form required	Timeframe
Autoimmune diarrhoea / colitis	Grade ≥ 2 diarrhoea persisting >48 hours and not suspected clinically or confirmed on lab testing to be due to infective causes	SAE report	Within 24 hours of confirmed diagnosis
Autoimmune uveitis	New or worsening inflammation of the eye, with ocular pain, erythema or blurring of vision.	SAE report	Within 24 hours of confirmed diagnosis
Autoimmune pneumonitis	Radiographic changes or pulmonary symptoms, unless due to infective causes	SAE report	Within 24 hours of confirmed diagnosis
Autoimmune hepatitis	Raised ALT, AST or bilirubin, unless due to infective causes or other concomitant medications	SAE report	Within 24 hours of confirmed diagnosis
Autoimmune dermatitis	New or worsening rash where non-autoimmune causes have been excluded	SAE report	Within 24 hours of confirmed diagnosis
Immune related endocrinopathy	New or worsening asymptomatic or symptomatic endocrine dysfunction	SAE report	Within 24 hours of confirmed diagnosis

Autoimmune renal toxicity	Raised creatinine unless due to nephrotoxicity of concomitant medications or outflow tract obstruction	SAE report	Within 24 hours of confirmed diagnosis
Autoimmune neurological toxicity	New or worsening radiographic changes or neurological symptoms	SAE report	Within 24 hours of confirmed diagnosis

**Cases of autoimmune toxicity must be reported on the trial-specific
SAE report by fax or email:
Fax: 0207 679 9861
Email: ctc.animate@ucl.ac.uk**

12.4.2 Potential Drug Induced Liver Injury (pDILI)

All occurrences of potential DILIs, meeting the defined criteria below, must be reported as SAEs.

Potential drug induced liver injuries should be reported from the first administration of nivolumab until five months post-treatment (or after this date if the investigator feels the event is related to nivolumab).

Event	Description	Form required	Timeframe
Potential drug induced liver injury (pDILI)	AST or ALT >3 x ULN and Total bilirubin >2 x ULN, without initial findings of cholestasis (elevated serum alkaline phosphatase) and No other immediately apparent causes of transaminase elevation and hyperbilirubinemia, including, but not limited to, viral hepatitis, pre-existing chronic or acute liver disease, or the administration of other drug(s) known to be hepatotoxic.	SAE report	Within 24 hours of diagnosis

**Cases of pDILI must be reported on the trial-specific
SAE report by fax or email:
Fax: 0207 679 9861
Email: ctc.animate@ucl.ac.uk**

12.4.3. Overdoses

Any overdoses of nivolumab, must be reported as both an SAE and an incident within 24 hours of the overdose occurring, regardless of whether the patient experiences any adverse events as a result of the overdose. Examples of potential overdoses are listed in the table below. If there is uncertainty about whether a patient has been overdosed, please contact the ANIMATE study team to discuss (ctc.animate@ucl.ac.uk).

Any adverse events occurring in patients who have received an overdose should be reported as serious events within the SAE report.

Event	Examples	Forms required	Timeframe
Overdose	- Administration of a nivolumab dose in excess of 240mg at any individual dosing episode - Administration of a cumulative nivolumab dose in excess of 1920mg (240mg x 8 doses)	- SAE report - Incident report	Within 24 hours of overdose occurring

**Overdoses must be reported by sending both the trial-specific
SAE report and incident report by fax or email:
Fax: 0207 679 9861
Email: ctc.animate@ucl.ac.uk**

12.4.4. Grade 3 or 4 infusion reactions to nivolumab

Any CTCAE grade 3 or 4 infusion reactions to nivolumab must be reported as SAEs.

Infusion reactions should be reported from first dose of nivolumab until completion of trial treatment.

See next page for reporting requirements.